ESNet: Anti-Hierarchical Encoder–Decoder Network With Surrounding Expansion and Stripping for Hyperspectral Image Classification

Zhaojie Pan, Genyun Sun, *Senior Member, IEEE*, Ziyan Ling, Aizhu Zhang, *Member, IEEE*, and Hang Fu

***Abstract*— Spatial information is crucial in deep spectral–spatial hyperspectral image (HSI) classification methods. Spatial features can be divided into central features and surrounding features, which have different significance in high-precision HSI classification tasks. However, the existing algorithms ignore the discrepancy between the two attributes, which weaken decisive characteristics and lose spatial structures. To resolve the above issues, a patch-based anti-hierarchical encoder–decoder network is proposed, namely expansion and stripping network (ESNet). First, a surrounding relation module (SRM) is proposed to change the patch scale. Unlike existing scale change modules, the SRM increases the scale in the encoder and decreases the scale in the decoder by controlling the expansion and stripping of peripheral pixels of the patch, which enables ESNet to amply exploit the critical central features while utilizing the surrounding information in the anti-hierarchical structure. Second, we design a spectral–spatial attention module (SSAM) to extract features. To better represent land cover attributes, SSAM employs large- and small-kernel convolutions to generate spatial, spectral, and spectral–spatial weighted features. Finally, a novel skip connection [deep shallow feature fusion module (DSFM)] is designed to promote the model to fuse deep and shallow features while preserving spectral sequences of patches. Combining DSFM with the encoder–decoder structure can more scientifically fuse deep semantics and shallow details. We conduct extensive experiments in six commonly used HSI datasets to demonstrate the superiority of the proposed model. The codes will be available from the website.**

***Index Terms*— 3-D spectral–spatial feature, anti-hierarchical encoder–decoder, central and surrounding feature, deep and shallow feature, hyperspectral image (HSI) classification.**

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Zhaojie Pan is with the School of Information and Control Engineering, Qingdao University of Technology, Qingdao 266520, China, and also with the School of Mathematics, Southeast Univeristy, Nanjing 211102, China (e-mail: panzhj99@163.com).

Genyun Sun is with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China, and also with the Laboratory for Marine Mineral Resources, Qingdao Marine Science and Technology Center, Qingdao 266237, China (e-mail: genyunsun@163.com).

Ziyan Ling is with the School of Information and Control Engineering, Qingdao University of Technology, Qingdao 266520, China (e-mail: lingziyan@163.com).

Aizhu Zhang is with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China, and also with the Key Laboratory of Poyang Lake Wetland and Watershed Research, Ministry of Education, Jiangxi Normal University, Nanchang 330022, China (e-mail: zhangaizhu789@163.com).

Hang Fu is with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: hangf_upc@163.com).

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I. Introduction

HYPERSPECTRAL image (HSI) classification has been exploited in diverse fields over the years, such as urban planning [1], precision agriculture [2], and tree species mapping [3]. Different from conventional remote sensing images, HSI can cover a much richer spectral range from the visible spectrum to far-infrared with dozens or even hundreds of continuous bands, providing detailed spectral information in addition to spatial relations [4]. This property allows researchers an opportunity to attribute each pixel of HSI to a certain category more precisely.

In past decades, various powerful algorithms have been proposed for HSI classification [5], [6], [7]. Earlier methods mainly focus on exploiting spectral features and resolving the issues resulting from the redundant spectral information. However, due to the influence of atmosphere, illumination, and sensor, the same material often has different spectra, called hyperspectral–heterospectral [8], which limits the performance of these spectral-based methods. Fortunately, some literature demonstrates that the spectral–spatial-based method can alleviate the hyperspectral–heterospectral problem [9].

In recent years, deep spectral–spatial methods have attracted extensive attention due to their ability to model abstract and deep spectral–spatial features [10], [11]. According to the structures of both input and output features, the deep spectral–spatial approaches are divided into two categories, that is, fully convolutional network (FCN)-based approaches [12], [13] and patch-based approaches [14]. The FCN-based methods can more adequately excavate global context and generate prediction images than patch-based methods, but the local features at different scales are usually not distinguished. Jia et al. [15] discovered that the features that play a decisive role in classification lie in the local, and the local features are more significant than global features. Moreover, the patch-based method can flexibly control the scale of spatial information by adjusting the patch size, which

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contributes to the design of specialized feature extraction structures based on the spatial structure of features for high-precision classification. Hence, these unique characteristics make the patch-based methods generally win more attention than FCN-based methods in deep spectral–spatial HSI classification. The development of patch-based methods can be divided into two categories: module-based methods and framework-based methods.

For module-based methods, researchers mainly propose or introduce several modules to exploit essential features. In [16], a variational autoencoder (VAE) is used to modify spectral sequential features by spatial features. In [17], a graph convolutional network (GCN) is introduced to enhance the representation of spatial and spectral information and obtain competitive results. Wang et al. [18] propose an RNN-based method called adaptive spectral–spatial multiscale network (ASSMN), which can perceive spectral multiscale information. In [19], [20], and [21], a vast amount of convolutional neural networks (CNNs)-based methods such as spectral–spatial residual network (SSRN), large kernel spectral and spatial attention networks (LKSSANs), and residual spectral–spatial attention network (RSSAN), have gained state-of-the-art (SOTA) results by introducing and developing powerful CNN structures. Recently, transformer-based methods have been developed. SpectralFormer [22] and spectral–spatial transformer network (SSTN) [23] are proposed successively to further improve the capability of excavating spectral and spatial features. While the above methods enhance the key feature representations and integrate spatial and spectral information, these methods mainly depend on the efficacy of the model component and neglect the exploration of the fine spatial structure.

For framework-based methods, researchers focus on designing an overall feature processing workflow to assist the feature extraction. An excellent framework can maximize the advantages of different feature extraction modules. The single branch is the most common structure, which exploits features and classifies images by cascading modules with different functions [24], [25]. However, the performance of the framework mainly depends on the feature extraction module. The two-branch framework can extract spectral and spatial information [26], multimodal [27] or multicontext [28] information by subnetworks. Compared with the two-branch framework, the multibranch framework can model more semantic expressions in parallel. In addition, the multibranch approach can extract multiscale features in a more convenient way, which contributes to the utilization of multiscale spatial representations in the mode [29]. Although these frameworks have different functions and obtain competitive performances, these frameworks are not developed based on geospatial attributes, which lead to their inability to adequately exploit fine-grained spatial information.

Recently, a few methods have been proposed for extracting fine geospatial features. In [30], a diverse region-based deep CNN model (DRCNN) is proposed to model spatial features in diverse local or global regions by multibranch. Yang et al. [31] propose an interactive learning framework (CSIL) to extract small-, medium-, and large-scale spatial surrounding information. DRCNN and CSIL effectively play the different functions of central and neighborhood features, and thus gain imposing results. However, these methods extract spatial information based on traditional multibranch frameworks, which make them inherently incapable of utilizing the framework to assist the model in mining different kinds of spatial information. In addition, these algorithms ignore a crucial geospatial attribute that central features are more significant than neighborhood features in spatial information.

To finely extract spatial information, we propose an anti-hierarchical encoder–decoder framework for HSI classification, termed "Expansion and Stripping Network" (ESNet), which can effectively utilize the framework to assist the modules in extracting fine-grained spatial information. Specifically, we first present a surrounding relation module (SRM) to change the scale of the patch. SRM enlarges the scale of the patch in the encoder and reduces the scale of the patch in the decoder, in this way the traditional encoder–decoder network forms the inverse hierarchy unique to this article. To enhance the feature extraction ability of ESNet, we design a spectral–spatial attention module (SSAM). SSAM can adaptively extract and fuse the spectral attributes, spatial context, and spectral–spatial associated information by long- and short-range 3-D attention. Ultimately, to scientifically fuse deep and shallow features, we design a deep shallow feature fusion module (DSFM) to integrate deep semantics and shallow details. The main contributions of this study are summarized as follows.

1) An anti-hierarchical encoder–decoder framework, called ESNet, is proposed to exploit fine-grained spatial information for HSI classification. To the best of our knowledge, ESNet is the first framework focused on the fine-grained application of geospatial information. ESNet purposefully integrates central and surrounding feature extraction with the encoder–decoder framework. In this way, the encoder can effectively model the multiscale spatial context of surrounding features, and the decoder can more explicitly emphasize the decisive characteristics of the central region.
2) To help the encoder and decoder extract features, we design SSAM to adaptively emphasize and fuse the spectral attributes, spatial contexts, and spectral–spatial relationships. Different from the existing attention methods, the SSAM employs large and small kernel convolution to generate 3-D weights, which enable SSAM to exploit long- and short-range spatial and spectral features.
3) To scientifically integrate deep and shallow features, a DSFM is designed to integrate abstract semantics and shallow details. The DSFM can adaptively merge deep and shallow features based on the discrepancy in the intrinsic properties of both shallow and deep features while preserving the spectral sequence of features.

The remainder of this article is organized as follows. Section II introduces related works. Section III describes the proposed ESNet in detail. In Section IV, experiments and results are presented and analyzed. Section V discusses the effectiveness and generalization of the structures of ESNet. The conclusions are summarized in Section VI.

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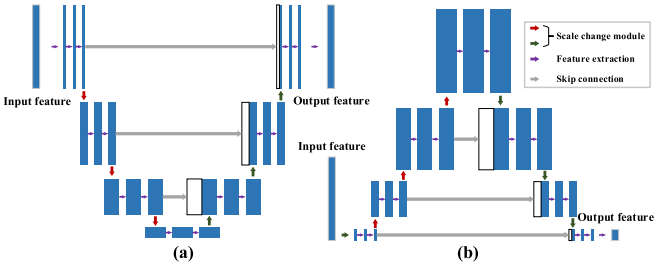


Fig. 1. Structures of (a) encoder–decoder and (b) anti-hierarchical encoder–decoder frameworks.

II. Related Works

A. Encoder–Decoder HSI Classification Methods

The encoder–decoder HSI classification methods are mostly inspired by UNet [32]. The UNet down-samples the features in the encoder and up-samples the features in the decoder. Fig. 1(a) shows a typical encoder–decoder structure. The encoder–decoder structure contains three core components: scale change module, feature extraction module, and skip connection. Researchers can design different feature extraction structures according to different tasks. The scale change module is located at the end of each stage of the encoder and decoder. The skip connection can fuse the low-level and high-level information. The feature extraction capability of the encoder–decoder framework is powerful, but it is less used in patch-based methods. In this article, we introduce the encoder–decoder into the patch-based HSI classification method and propose an anti-hierarchical encoder–decoder classification framework for modeling center and neighborhood spatial information. Fig. 1(b) shows an anti-hierarchical encoder–decoder structure. There are two main differences between the traditional structure and the anti-hierarchical structure: (1) the anti-hierarchical framework expands the feature scale in the encoder and reduces the feature scale in the decoder, and (2) the scales of input and output features in the anti-hierarchical framework are different. In Section III, we will present the detailed application of the anti-hierarchical structure in patch-based HSI classification.

B. Attention and Its Application to HSI Classification

Attention can focus on significant information autonomously employing feature weighting and thus has a wide range of applications in the HSI. Early attention structures, including squeeze-and-excitation networks (SENets) [33] and convolutional block attention modules (CBAMs) [34], use convolution operations or pooling functions to process features to generate weight maps. Some methods introduce these attention structures to the enhancement of spectral and spatial information and obtain excellent performance [35]. Then, self-attention [36] was developed to calculate the feature weights by the covariance between the same features in different eigenspaces. The self-attention-based methods were successfully employed to explore the long-range spectral and spatial correlation. In [37], the transformer was proposed to model long-range attributes by multihead self-attention. Many transformer-based HSI classification methods have been proposed and achieve imposing results [38], [39]. However, the existing attention structure makes it hard to simultaneously extract and fuse long- and short-range spectral–spatial integration features, so we design a novel attention structure to exploit spectral, spatial, and spectral–spatial relation features.

III. Proposed ESNet

Surrounding information and center information play different roles in HSI classification. The center information is more crucial in characterizing the center pixel than the surrounding information. In this article, we propose an anti-hierarchical encoder–decoder framework, called ESNet, as shown in Fig. 2. ESNet consists of three core structures: SRM, SSAM, and DSFM. SRM can control the involvement of multiscale surrounding information in feature extraction of the encoder and decoder. As a result, the model can facilitate the utilization of surrounding information without inhibiting the expression of the deterministic central features. SSAM is a new attention structure and can adoptively mine and fuse long- and short-range spectral, spatial, and spectral–spatial relation information by large- and small-kernel spectral and spatial convolutions. The DSFM is developed to fuse deep and shallow information. DSFM deals with deep and shallow information with the same spatial coverage scale through spectral dimension adaptive weighting, which is more consistent with the HSI. In this section, we will describe the core modules and the overall classification process of ESNet in detail.

A. SRM for Changing Image Scale

Spatial context information from the central and surrounding regions is crucial for improving HSI classification accuracy, and the contributions of these features are not equally critical [31]. In this section, we propose SRM to assist the model in extracting center and surrounding features in a differentiated and focused manner. The three subarchitectures of the SRM, that is, SRMa, SRMb, and SRMc are illustrated in Fig. 3.

Suppose d is the number of stages of the encoder, and the initial input feature of SRM is $F \in R^{c\times(2r+1)\times(2r+1)}$, where c denotes the number of the first several principal components (PCs) selected by PCA, and r is the radius of the spatial receptive field of the patch. In SRM, the cheap channel attention function (CCA) is first proposed to change the channel scale of the feature, and the CCA can be expressed by the following formula:

$$F' = \mathrm{con}_{1\times1}(\mathrm{CAM}(F)) \tag{1}$$

$$\mathrm{CAM}(F) = F + f_{\mathrm{re}}(F) \otimes \mathrm{soft\,max}(f_{\mathrm{re}}(F) \otimes f_{\mathrm{tr}}(f_{\mathrm{re}}(F))) \tag{2}$$

where $\mathrm{con}_{1\times1}(\cdot)$ is channel convolution, $\mathrm{CAM}(\cdot)$ denotes channel attention module [36], $f_{\mathrm{re}}(\cdot)$ and $f_{\mathrm{tr}}(\cdot)$ denote reshape and transpose function, respectively.

In the first stage of encoder, SRMa generates the initial center feature $F_{E,1}^{\mathrm{SRM}}$ by striping $\lfloor r/\varepsilon \rfloor (d-1)$ edges of $F'_{E,1}$, where $\lfloor\cdot\rfloor$ is round function and $\varepsilon, d \le \varepsilon \le r$ is the scale factor. The following relationship exists between $F_{E,1}^{\mathrm{SRM}}$ and $F'_{E,1}$:

$$F_{E,1}^{\mathrm{SRM}}(x,y) = F'_{E,1}\big(x+\lambda(d-1), y+\lambda(d-1)\big) \tag{3}$$

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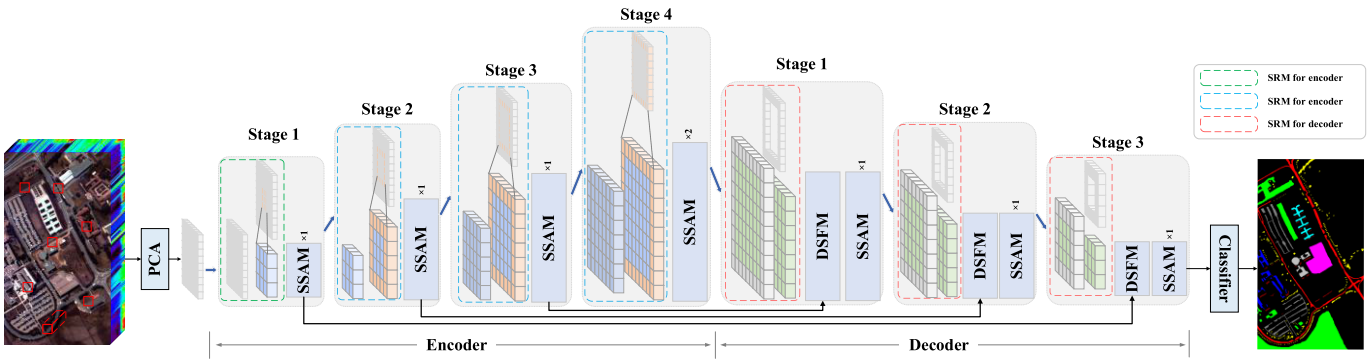


Fig. 2. Overview of the proposed of ESNet. The framework of ESNet involves an Encoder and a Decoder task, and SRM, SSAM, and DSFM are the newly proposed approaches. Different from the typical encoder–decoder framework, in this work, the encoder expands the spatial scale of the patch, while the decoder condenses the scale. In each stage, SRM first is employed to change the feature scale. The SSAM is used to exploit features and the number of SSAM in different stages is different. The DSFM is only located in the decoder and integrates the deep and shallow features with the same spatial scale. Finally, the classifier fuses the 3-D output features of the decoder as 1-D features and labels the center pixel of the patch.

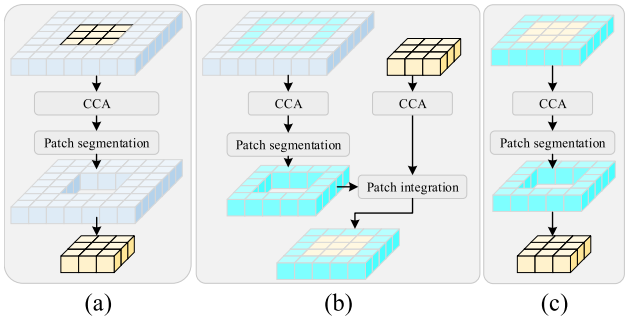


Fig. 3. Sub-architectures of SRM. (a) SRMa and (b) SRMb are employed by encoder. (c) SRMc is used by decoder.

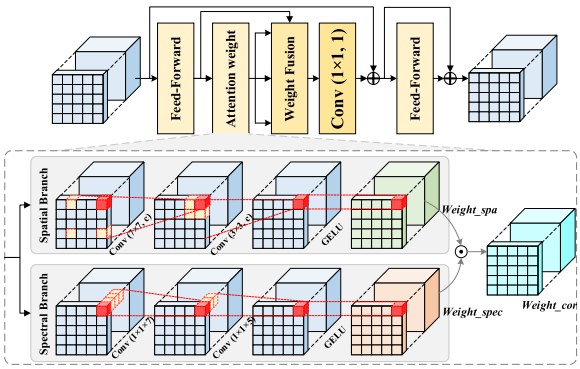


Fig. 4. Structure of the proposed SSAM. The colored grids represent the location of convolution kernel, and the red grid means the center point.

where $\lambda = \lfloor r/\varepsilon \rfloor$, x and y are the row and column coordinates, respectively. x and y vary in the set $\{t|0 \le t < 2(r-\lambda(d-1))+1, t \in Z\}$; then, the $F_{E,1}^{\mathrm{SRM}}$ is fed into SSAM for exploiting critical information, and $F'_{E,1}$ is preserved for the SRM of the second stage of encoder.

In the ith stage of encoder, SRMb has two input features, $F_{E,i-1}^{\mathrm{out}}$ and $F'_{E,i-1}$. SRMb first feeds the two input features into CCA to generate $F_{E,i}$ and $F'_{E,i}$, and then the new center feature $F_{E,i}^{\mathrm{SRM}}$ is generated by combining the center feature $F_{E,i}$ with the surrounding feature $F'_{E,i}$. As shown in Fig. 3(b), when the radius of the $F_{E,i}$ is 1, the cyan pixel block in $F'_{E,i}$ will be combined with $F_{E,i}$, and the radius of the generated new center feature $F_{E,i}^{\mathrm{SRM}}$ is 2. Hence, the new central feature $F_{E,i}^{\mathrm{SRM}}$ generated by SRMb meets the following conditions:

$$F_{E,i}^{\mathrm{SRM}}(x,y) = \begin{cases} F_{E,i}(x-\lambda, y-\lambda) \\ F'_{E,i}(r-r_{E,i}-\lambda+x, r-r_{E,i}-\lambda+y) \end{cases} \tag{4}$$

where $r_{E,i}$ is the radius of $F_{E,i}$, x and y both belong to the set $\{t|0 \le t < 2(r_{E,i}+\lambda)+1, t \in Z\}$. According to (3) and (4), we can speculate that combining SRM with the encoder not only allows more surrounding information to be involved in the exploitation of the central feature but also effectively limits the distraction that is far from the central pixel.

Peripheral information mainly plays an auxiliary role in the final classification, so the decoder adopts SRMc to reduce the interference of surrounding information to the central region. In the ith stage of the decoder, the following relationship exists between the new central feature $F_{D,i}^{\mathrm{SRM}}$ and output feature $F_{D,i}$ generated by the CCA:

$$F_{D,i}^{\mathrm{SRM}}(x,y) = F_{D,i}(x+\lambda, y+\lambda) \tag{5}$$

where $F_{D,i} = \mathrm{con}_{1\times1}(\mathrm{CAM}(F_{D,i-1}^{\mathrm{out}}))$, $F_{D,i-1}^{\mathrm{out}}$ is the output of $i-1$th stage of the decoder, when i is 1, the $F_{D,i-1}^{\mathrm{out}}$ is $F_{E,d}^{\mathrm{out}}$. x and y both vary in the set $\{t|0 \le t < 2(r_{D,i}-\lambda)+1, t \in Z\}$, where $r_{D,i}$ is the radius of $F_{D,i}$.

B. Spectral–Spatial Attention Module

The spectral and spatial features are essential for high-precision classification tasks. To efficiently utilize spectral information, spatial information, and spectral–spatial correlation, we designed SSAM that can exploit more representative and more meaningful features from HSI, as shown in Fig. 4. The input feature of SSAM is F^{SRM} or F^{DSFF}. To model the spectral dependency, SSAM first feeds the input feature into feed-forward [40] to obtain the central feature F_c^{FF}, and then F_c^{FF} is fed into attention structure to construct 3-D spatial, spectral, and spectral–spatial correlation weights.

The attention structure contains two branches, that is, spectral branch and spatial branch. In the spatial branch, we first use large kernel depthwise dilation convolution to aggregate

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long-range spatial information. Although dilated convolution can effectively improve the receptive field, it is easy to lose details. Therefore, we then use small kernel convolution to exploit the local context. Finally, the Gaussian error linear unit (GELU) [41] is used to obtain a spatial weight map. It is expressed by the following formula:

$$W_{\text{spa}} = \text{GELU}\left(f_{\text{con}(3\times3,c,1)}\left(f_{\text{con}(7\times7,c,3)}\left(F_c^{\text{FF}}\right)\right)\right) \tag{6}$$

where $\text{GELU}(\cdot)$ is the activation function and $f_{\text{con}(k\times k,g,\text{di})}(\cdot)$ is 2-D convolution, where k is the size of the kernel, g is the group that is used to control the number of channels for information aggregation, and di is dilation rate defaults to 1.

In the spectral branch, we use 3-D convolution to generate a spectral weight map, which is calculated in the following way:

$$W_{\text{spec}} = \text{GELU}\left(f_{\text{con}(1\times1\times5)}\left(f_{\text{con}(1\times1\times7)}\left(F_c^{\text{FF}}\right)\right)\right) \tag{7}$$

where $f_{\text{con}(k\times k\times l)}$ represents 3-D convolution, $k \times k \times l$ is the size of the convolution kernel. We can see from (7) that we, respectively, set the convolution kernel size of the two spectral dimensions to 7 and 5 to make effective use of long- and short-range spectral correlations. In addition, we also set the sizes of the convolution kernel of spatial dimension to 1 to avoid interference with spatial interaction calculation.

Since HSI is a 3-D structure, spectral information of ground objects is intrinsically correlated with other ground objects in its surroundings. Therefore, after obtaining spectral and spatial weight maps, a joint spectral–spatial weight map W_{cor} can be obtained by the following formula:

$$W_{\text{cor}} = W_{\text{spa}} \odot W_{\text{spec}} \tag{8}$$

where $\odot$ represents the dot product, which is the multiplication of the corresponding elements of the matrix. Then, three weighted feature maps are obtained by

$$\begin{cases} F_{c,\text{spa}} = F_c^{\text{FF}} \odot W_{\text{spa}} \\ F_{c,\text{spec}} = F_c^{\text{FF}} \odot W_{\text{spec}} \\ F_{c,\text{cor}} = F_c^{\text{FF}} \odot W_{\text{cor}}. \end{cases} \tag{9}$$

It is difficult to measure the attribution of spectral, spatial, and spectral–spatial correlation on the final classification results, so we carry out a weighted fusion of these three feature maps. Then, the final weighted feature F_c^W can be obtained by

$$F_c^W = f_{\text{con}(1\times1,1,1)}\left(\alpha \times \left(F_{c,\text{spa}}\right) + \beta \times \left(F_{c,\text{spec}}\right) + \gamma \times \left(F_{c,\text{cor}}\right)\right) \tag{10}$$

where α, β, and γ are weighting parameters in the range of [0, 1], which can be continuously optimized along with network training to achieve the best fusion of weighted features. Finally, two residual connections and a feed-forward are set to get the output of SSAM F^{SSAM}. It can be written as

$$F^{\text{SSAM}} = \left(F_c^W + F^{\text{SRM}}\right) + \text{FF}\left(F_c^W + F^{\text{SRM}}\right) \tag{11}$$

where the input feature of SSAM is F^{SRM}. Note that in the stage of encoder or decoder, the input feature of the second and subsequent SSAM is the output feature of the previous SSAM, and the output of the stage F^{out} is the output of last SSAM. Unlike traditional attention structures, SSAM adopts large- and small-kernel convolution to generate three kinds of weight maps, which enable ESNet to more efficiently use spectral, spatial, and spectral–spatial correlation information.

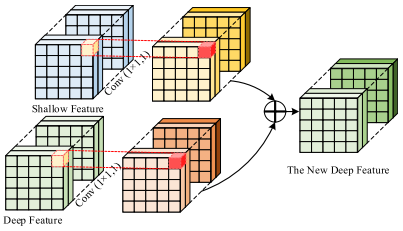


Fig. 5. Structure of the proposed DSFM.

C. Deep Shallow Feature Fusion

Deep and shallow feature fusion needs to consider the characteristics of 3-D patches in spatial and spectral dimensions. In encoder and decoder, the spatial information of the patch in the deeper layer can effectively represent the larger receptive field and more abstract semantics after continuous aggregation and transformation. However, based on the operating principle of convolution, we know that the core representation of each pixel in the spatial dimension is still the feature of this pixel. In addition, the effective use of feature spectral sequences can effectively improve the performance of HSI classification models [42], [43], [44], but the traditional skip connection destroys the sequence information. Therefore, we propose DSFM to integrate effectively deep and shallow features, and the structure of DSFM is shown in Fig. 5.

In the jth stage of the decoder, the input of DSFM is the output of the $d-j$th stage of the encoder $F_{E,d-j}^{\text{out}} = [A_1, A_2, \ldots, A_k, \ldots, A_{c_{D,j}}]$ and the output of the SRMc $F_{D,j}^{\text{SRM}} = [B_1, B_2, \ldots, B_k, \ldots, B_{c_{D,j}}]$, where $A_k, B_k \in R^{(2r_{D,j}+1)\times(2r_{D,j}+1)}$, $c_{D,j}$, and $r_{D,j}$ represent the number of channels and the size of radius, respectively. The output feature of DSFM $F_{D,j}^{\text{DSFF}} = [C_1, C_2, \ldots, C_k, \ldots, C_{c_{D,j}}]$ is obtained by

$$C_k = \alpha_{k,j} \times A_k + \beta_{k,j} \times B_k \tag{12}$$

where α_k and β_k are weighting parameters in the range of [0, 1], which can be trained.

D. ESNet for HSI Classification

ESNet is a patch-based classification model, that is, the input of ESNet is 3-D patches, and its output is 1-D sequences. Based on the description of Section II-A, we know that the size of the output of the decoder is smaller than that of the encoder. As a result, the classifier can integrate the output of the decoder more conveniently. The classifier can be expressed as follows:

$$\begin{aligned} p &= \text{FCN}\left(\text{mean}\left(F_{D,d-1}^{\text{out}}\right)\right) \\ Y &= \arg\max(p) \end{aligned} \tag{13}$$

where $\text{mean}(\cdot)$ indicates that the information in the spatial dimension of the feature is averaged by the mean function, $\text{FCN}(\cdot)$ is fully connected, $\arg\max(\cdot)$ is the argmax function. $p = [p_0, p_1, \ldots, p_k, \ldots, p_K]$, where p_k is the prediction

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Algorithm 1 Pseudocode of the Proposed ESNet

Input: HSI X, Indexes of samples x_{in}, the number of PCs c, the radius of patch r, the depth of encoder d, the scale factor λ, the number of SSAM in each stage of encoder and decoder $s = [s_1, s_2, \ldots, s_{2d-1}]$.
Output: Labels of samples Y.
Data preprocessing:
Based X, c, x_{in} and r, obtain the initial input feature $F \in R^{c\times(2r+1)\times(2r+1)}$ by PCA and patch segmentation.
Encoder:
For $i = 1$ **to** d **do**
If $i = 1$: Feed F into SRMa to generate initial central feature $F_{E,1}^{SRM}$ and $F'_{E,1}$.
Else: Feed F'_{i-1} and $F_{E,i-1}^{out}$ into SRMb to generate central feature $F_{E,i}^{SRM}$ and $F'_{E,i}$.
For $j = 1$ **to** s_i **do**
If $j = 1$: Feed $F_{E,i}^{SRM}$ into SSAM to obtain $F_{E,i}^{SSAM}$.
Else: Update $F_{E,i}^{SSAM}$ by SSAM.
End
End
Decoder:
For $i = 1$ **to** $d - 1$ **do**
If $i = 1$: Feed $F_{E,d}^{out}$ into SRMc to generates $F_{D,1}^{SRM}$.
Else: Feed $F_{D,i-1}^{out}$ into SRMc to generates $F_{D,i}^{SRM}$.
Feed $F_{D,i}^{SRM}$ and $F_{E,d-i}^{out}$ into DSFM to generates $F_{D,i}^{DSFF}$.
For $j = 1$ **to** s_{d+i} **do**
If $j = 1$: Feed $F_{D,i}^{DSFF}$ into SSAM to obtain $F_{D,i}^{SSAM}$.
Else: Update $F_{D,i}^{SSAM}$ by SSAM.
End
Classifier:
Feed $F_{D,d-1}^{out}$ into (13) to obtain Y.

probability that the central pixel of the 3-D patch belongs to the category k.

In addition, we employ the focal loss [45] to estimate the prediction error for optimal ESNet. After that, Y represents the class of the center pixel. In the ESNet, the feature map is not directly processed by the spatial convolution but interacts with the surrounding information by constructing weights, which allow for maximum preservation of the essential features of each pixel block when using the surrounding information. A pseudocode of ESNet is given in Algorithm 1.

IV. Experiment and Result

In this section, we first introduce six HSI datasets and then describe the experimental evaluation metrics, SOTA DL-based methods for comparison, and hardware configuration. To analyze the performance of ESNet, we test the influence of related parameters on the accuracy of the algorithm, and finally, we qualitatively and quantitatively analyze the results of the proposed ESNet and SOTA methods in the HSI datasets.

A. Datasets Description

1) Spatial Disjoint Dataset: We used three well-known spatially disjoint datasets, and the number of training and testing samples of the three datasets are shown in Table I.

TABLE I
Numbers of Training and Test Samples Used in IP, PU, and UH2013 Datasets

IP dataset						PU dataset			UH2013 dataset					
Class No.	Train	Test	Class No.	Train	Test	Class No.	Train	Test	Class No.	Train	Test	Class No.	Train	Test
1	29	25	10	470	503	1	548	6304	1	198	1053	10	191	1036
2	762	675	11	1414	1065	2	540	18146	2	190	1064	11	181	1054
3	435	404	12	328	282	3	392	1815	3	192	505	12	192	1041
4	146	99	13	132	80	4	524	2912	4	188	1056	13	184	285
5	232	274	14	728	545	5	265	1113	5	186	1056	14	181	247
6	394	354	15	291	99	6	532	4572	6	182	143	15	187	473
7	16	12	16	57	44	7	375	981	7	196	1072			
8	235	250				8	514	3364	8	191	1053			
9	10	10				9	231	795	9	193	1059			

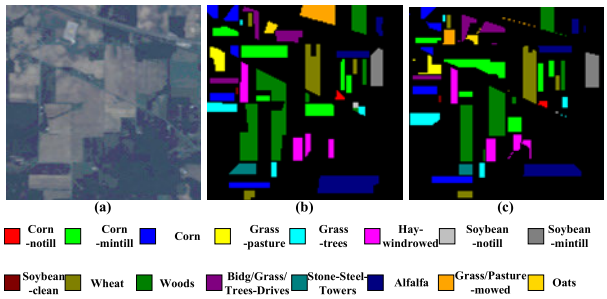


Fig. 6. IP dataset. (a) True-color image of IP. Spatial distribution of (b) training and (c) testing samples of IP.

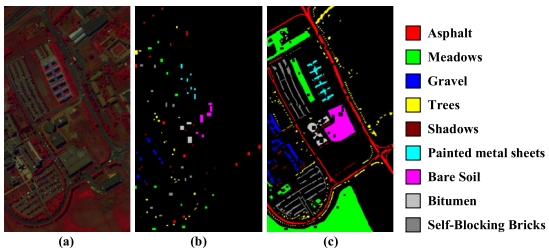


Fig. 7. PU dataset. (a) False-color image of PU. Spatial distribution of (b) training and (c) testing samples of PU.

The details of these datasets are shown below. *Indian Pines dataset (IP):* IP contains consists of 145 × 145 pixels, 20 m per pixel resolution, and 200 spectral bands from 400 to 2500 nm. This dataset contains 10 400 labeling pixels with 16 ground-truth classes. The specific ground features of the IP dataset are shown in Fig. 6. *Pavia University dataset (PU):* PU was acquired by using the Reflective Optics Spectrographic Imaging System (ROSIS) sensor over the area of Pavia University (PU), Northern Italy. The whole image has nine classes and 610 × 340 pixels, with a spatial resolution of 1.3 m. The number of wavelength bands is 103 ranging from 430 to 860 nm. The specific ground features of the PU dataset are shown in Fig. 7. *Houston 2013 Data (UH2013):* The UH image was gathered by an ITRES CASI-1500 sensor, which covers the area of the University of Houston. The HSI comprises 349 × 1905 pixels with 144 wavelength bands in the range of 0.38–1.05 μm, with a spatial resolution of 2.5 m. As shown in the Fig. 8, the UH dataset contains 15 different classes.

2) Few-Shot Dataset: To validate the performance of the ESNet in insufficient and imbalanced sample problems, we used three datasets with more data volumes and more

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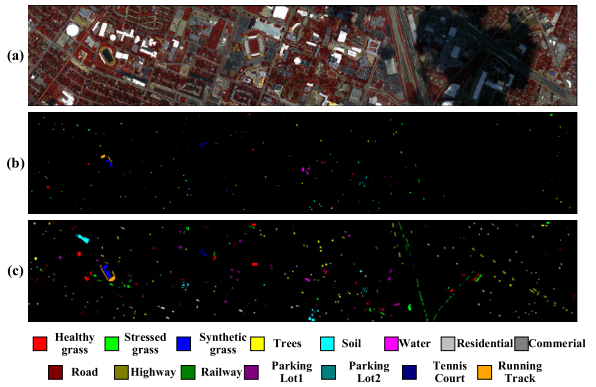


Fig. 8. UH2013 dataset. (a) False-color image of UH2013. Spatial distribution of (b) training and (c) testing samples of UH2013.

TABLE II

NUMBERS OF TRAINING AND TEST SAMPLES USED IN UH2018, WH, AND TREE DATASETS

UH2018 dataset						WH dataset						TREE dataset		
Class No.	Train	Test	Class No.	Train	Test	Class No.	Train	Test	Class No.	Train	Test	Class No.	Train	Test
1	95	9400	13	468	46315	1	15	14026	13	23	22484	1	81	80662
2	320	31610	14	99	9783	2	4	3508	14	8	7348	2	6	5334
3	7	637	15	70	6857	3	22	21799	15	2	1000	3	4	4171
4	135	13275	16	115	11322	4	164	16312	16	8	7254	4	15	14916
5	50	4878	17	2	137	5	7	6211	17	4	3006	5	41	40234
6	45	4438	18	67	6538	6	45	44512	18	4	3213	6	26	25435
7	3	235	19	53	5176	7	25	24078	19	9	8703	7	3	2997
8	396	39109	20	68	6722	8	5	4049	20	4	3482	8	9	8315
9	2240	22175				9	11	10808	21	2	1326	9	11	10730
10	450	44494				10	13	12381	22	5	4035	10	11	10461
11	322	31811				11	12	11003				11	11	10159
12	16	1501				12	9	8945				12	1	67

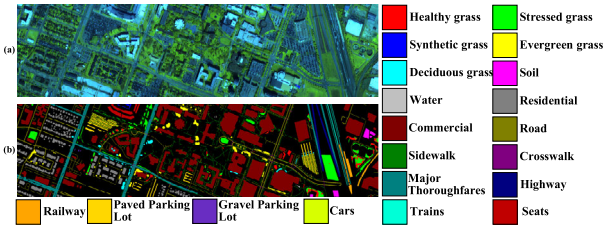


Fig. 9. UH2018 dataset. (a) False-color image of UH2018. (b) Spatial distribution of ground truth of UH2018.

complex scenarios, and the sample matches used for the experiments conducted in these three datasets are shown in Table II. The details of these three datasets are described as follows. *Houston 2018 dataset (UH2018):* UH2018 was captured by the ITRES CASI 1500 sensor over the University of Houston and contains 501 015 labeling pixels with 20 ground-truth classes. This database contains consists of 601 × 2385 pixels, 1 m ground sampling distance (GSD), and 48 spectral bands from 380 to 1050 nm. The specific ground features of the UH2018 dataset are shown in Fig. 9. *WHU-Hi-HongHu dataset (WH):* The WH image was recorded by the Headwall Nano-Hyperspec sensor with fragmented plots and various crops [13]. The image size in pixels is 940 × 475, with a spatial resolution of 0.043 m, composed of 270 bands ranging from 400 to 1000 nm. The ground truth of this scene consists of 22 classes. A true-color image and the ground-truth map are shown in Fig. 10. *Forest farm dataset (TREE):* The image was gathered by the AISA Eagle II sensor, which covers the area of Jiepai, Gaofeng forest farm, Nanning city of Guangxi [46].

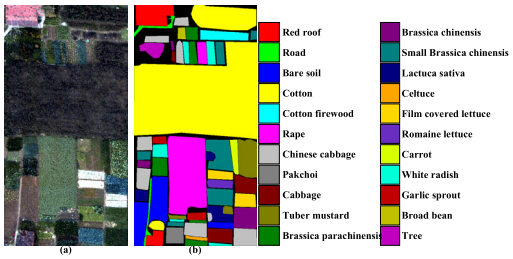


Fig. 10. WH dataset. (a) True-color image of WH. (b) Spatial distribution of ground truth of WH.

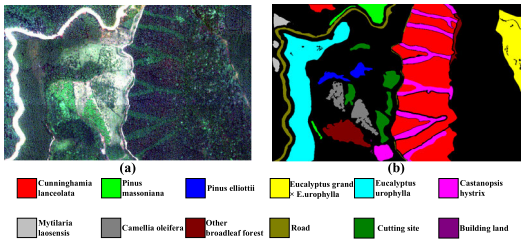


Fig. 11. TREE dataset. (a) True-color image of TREE. (b) Spatial distribution of ground truth of TREE.

The HSI cube comprises 1096 × 715 pixels with a resolution of 0.2 m per pixel, and it contains 125 wavelength bands in the range of 400–1000 nm. As shown in Fig. 11, there exist 12 different classes.

B. Experimental Setup

1) Evaluation Metrics: The classification maps obtained by different models are visualized to make a qualitative comparison. To quantify the classification performance of ESNet, we use six metrics: 1) the overall accuracy (OA); 2) average accuracy (AA); 3) the kappa coefficient (kappa); 4) the producer's accuracies (PA) of each land-cover category; 5) time (T) taken by the model to train one epoch and test; and 6) statistical significance (S) based on McNemar's test [47]. In McNemar's test, $S > 1.96$ indicates that the improvement of accuracies yielded by the benchmark model is statistically significant as compared with other methods. In this article, the benchmark model is ESNet.

2) For Hyperparameters: To explore the performance boundary of ESNet, five kinds of hyperparameter tuning experiments were conducted. First, we analyze the number of PCs and patch sizes in six datasets to examine the effect of sizes of input features on the model performance. Then, we evaluate two parameters related to the structure of ESNet: the encoder depth and the scale factor. Finally, we analyze the effect of the combinations of large- and small-range convolution kernels in SSAM on the feature extraction capability of the model. Considering the existence of duplicate scenarios for the six datasets, the last three hyperparameters are estimated in the UH2018, WH, and TREE datasets.

3) Comparison With SOTA Backbone Networks: To analyze the effectiveness of ESNet, we compare ESNet with other existing SOTA methods based on various types and structures. The comparison algorithms are divided into two categories, transformer-based, and other method-based methods. For the

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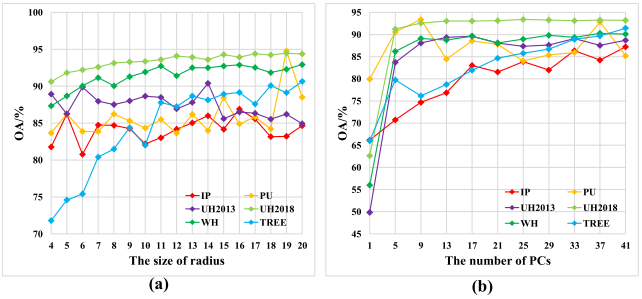


Fig. 12. Effects of different (a) size of patch radius and (b) number of PCs for ESNet on the IP, PU, UH2013, UH2018, WH, and TREE datasets.

transformer-based methods, we choose SSTN [23], spectral–spatial feature tokenization transformer (SSFTT) [38], hyperspectral image transformer (HiT) [48], and local transformer with spatial partition restore network (SPRLT-Net) [39] to analyze the effectiveness of the ESNet. For the other method-based models, we use RNN-based ASSMN [18], GCN-based fast dynamic graph convolutional network (FDGC) [17], attention-based adaptive spectral–spatial kernel improved residual network (A2S2K) [49], and LKSSANs [20].

4) Implementation Details: All the experiments are implemented on Windows 10 and an 8-GB GPU of Nvidia GeForce GTX 1080. The comparison algorithms adopt their optimal parameters as described in the corresponding references. The proposed networks are carried out using the PyTorch platform as the backend. We train the network for 200 epochs to adopt the Adam optimizer [50] with a batch size of 28 and a learning rate of 0.005. To avoid biased estimation, all experiments are conducted with five independent tests, and the average values are reported for all the evaluation metrics.

C. Hyperparameter Analysis for ESNet

To analyze the effect of the patch size on the model performance, we set the size of the radius of the patch from 4 to 20 for the six datasets, and the evolution of OA along with the radius of the patch is shown in Fig. 12(a). In Fig. 12(a), we see that each OA is increasing as the patch size rises until the radius is more than 15 in UH2018, WH, and TREE datasets. In the IP and UH2013 datasets, OA varies up and down around the accuracy at 8. In the PU dataset, the OA shows a wavy rise and attains its maximum value at 19. Hence, we finally set the radius to 15 in the UH2018, WH, and TREE datasets and set the radius to 19 in the PU dataset, respectively. In the IP and UH2013 datasets, the radius is set to 8.

As for the number of PCs, we set it to range from 1 to 41 with an interval of 4. Fig. 12(b) displays the OA versus the number of PCs in six datasets. It can be observed that the OA varies less when the number of PCs exceeds 9 in the UH2018 and WH datasets, the OA tends to stabilize when the number exceeds 17 in the UH2013 datasets, and the OA is highest at 9 in the PU dataset and at 41 in TREE and IP datasets. Hence, considering model performance and complexity, the number of PCs is given as 9 in PU, UH2018, and WH datasets, and the number is set to 41 in IP and TREE datasets and to 17 in the UH2013 dataset.

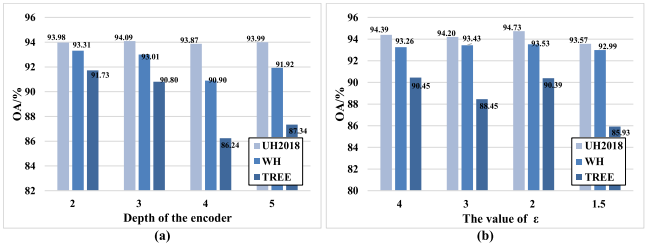


Fig. 13. Effect of different (a) depth and (b) value of ε for ESNet on the UH2013, WH, and TREE datasets.

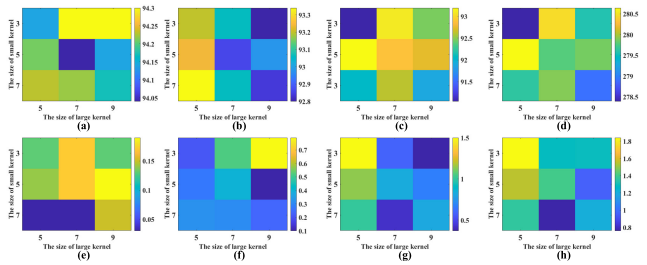


Fig. 14. Effect of combination of size of large and small kernel in the UH2018, WH, and TREE dataset in five experiments. (a)–(c) Mean of OA. (e)–(g) Standard deviation of OA. (d) and (h) Sum of mean and standard deviation.

The value of ε and the depth of the encoder are combined to restructure ESNet for exploiting the center and surrounding features. We set the depth of the encoder to the range of {2, 3, 4, 5} to analyze the effect of the depth on the model performance, and the deeper the depth, the more delicate the use of center and surrounding information. The results of depth tuning are reported in Fig. 13(a). As we can see, ESNet is less sensitive to the change in depth at the UH2018 and WH datasets, but the OA of the model at the TREE dataset slightly decreases with increasing depth. Therefore, we finally set the depth of the encoder to 2 for all the datasets.

When the encoder depth is 2, an experiment is conducted to discuss the effect of the number of layers of expand or strip edge pixels per step, which is controlled by ε. As indicated in Fig. 13(b), with the decrease of ε in the range of {4, 3, 2, 1.5}, OA initially increases, then slightly decreases, and obtains optimal results at a value of 2 in UH2018 and WH datasets. Although the optimal result in the TREE dataset is at 4, the discrepancy with the result at 2 is tiny. Therefore, considering the OA of ESNet on each dataset, the model has the best performance when the value of ε is set at 2 for all datasets.

We set the size of large and small kernels to the range of {5, 7, 9} and {3, 5, 7}, respectively. The large kernel convolution denotes dilated convolution, so the range of variation of the large kernel is equivalent to {9, 13, 17}. The means and standard deviations of OA are shown in Fig. 14. According to the results, the means of OA are better at (7, 3) and (5, 5), and the standard deviations of OA are better at (7, 3) and (9, 3). Therefore, we set the combinations of large- and small-range convolution kernels to (7, 3) for all the datasets.

Based on the above results, we can find that the ESNet is more suitable for inputs with smaller spatial sizes in images with larger GSD. In the channel dimension, the performance

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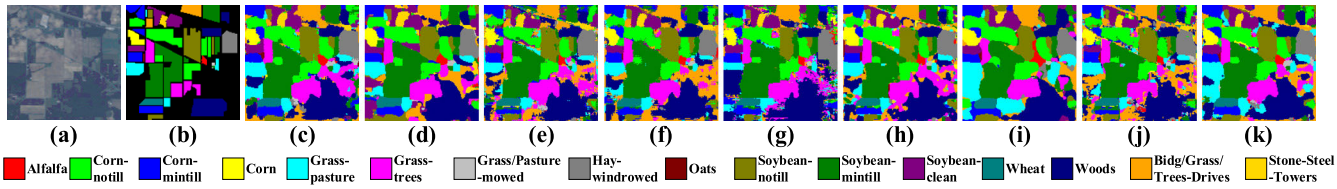


Fig. 15. Visualization of the classification maps for the IP dataset. (a) True-color image of IP. (b) Ground truth of IP. (c) SSTN. (d) SSFTT. (e) HiT. (f) SPRLT. (g) ASSMN. (h) A2S2K. (i) FDGC. (j) LKSSAN. (k) ESNet.

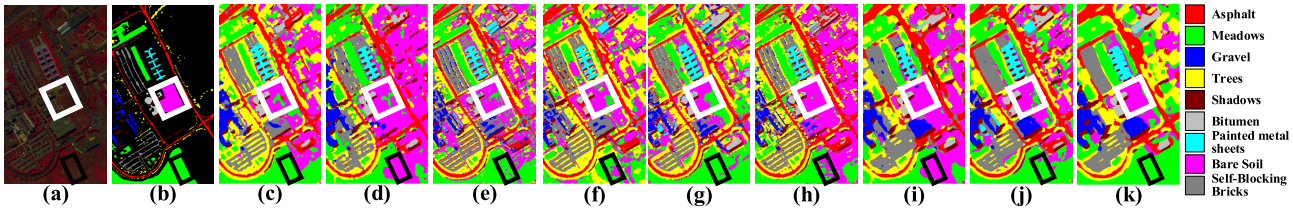


Fig. 16. Visualization of the classification maps for the PU dataset. (a) True-color image of PU. (b) Ground truth of PU. (c) SSTN. (d) SSFTT. (e) HiT. (f) SPRLT. (g) ASSMN. (h) A2S2K. (i) FDGC. (j) LKSSAN. (k) ESNet.

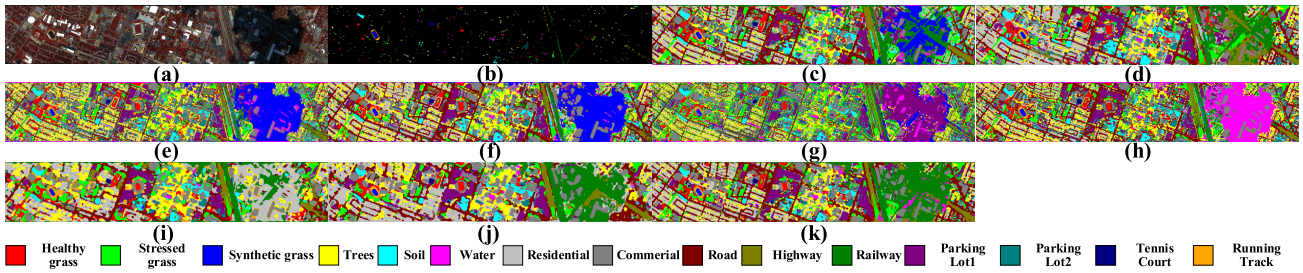


Fig. 17. Visualization of the classification maps for the UH2013 dataset. (a) False-color image of UH2013. (b) Ground truth of UH2013. (c) SSTN. (d) SSFTT. (e) HiT. (f) SPRLT. (g) ASSMN. (h) A2S2K. (i) FDGC. (j) LKSSAN. (k) ESNet.

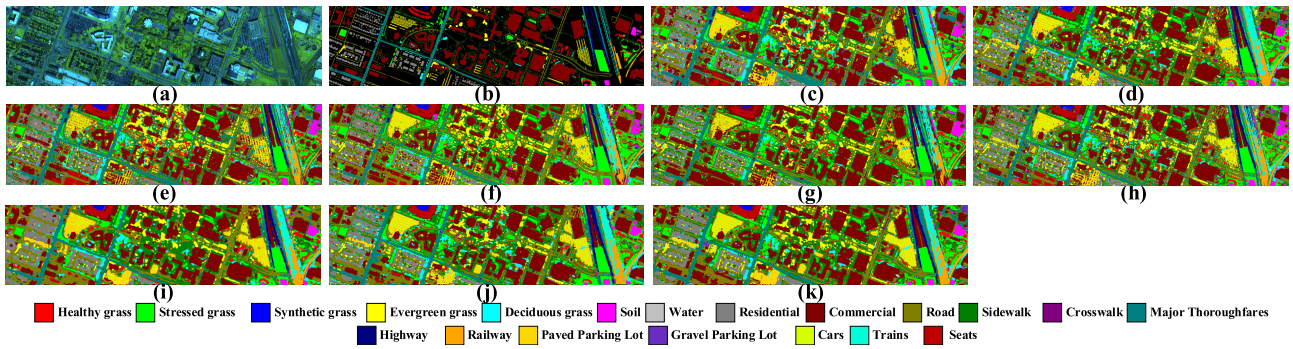


Fig. 18. Visualization of the classification maps for the UH2018 dataset. (a) True-color image of UH2018. (b) Ground truth of UH2018. (c) SSTN. (d) SSFTT. (e) HiT. (f) SPRLT. (g) ASSMN. (h) A2S2K. (i) FDGC. (j) LKSSAN. (k) ESNet.

of the ESNet no longer rises when the PC is bigger than a fixed value, and this value is different in different images. Moreover, we can also speculate that more and finer surrounding information does not equal better model performance but degrades the model performance. This result indirectly proves that center information is more significant than surrounding information.

D. Classification Results for ESNet

In this section, we demonstrate the effectiveness of our approach by qualitatively and quantitatively comparing ESNet with a series of SOTA algorithms in six datasets with different spatial resolutions and scenarios. Moreover, we analyze the results of each model in different training samples in the six datasets. Finally, we evaluate the complexity of ESNet in UH2018, WH, and TREE datasets with larger data volumes. Figs. 15–20 show the classification maps obtained by each algorithm, and the evolution of OA along with the number of training samples on different datasets is shown in Fig. 21. The quantitative results obtained by different methods in different datasets are summarized in Tables III–VII.

1) Quantitative Evaluation: The quantitative results of OA, AA, and Kappa of each model in spatial disjoin datasets are shown in Table III. We can see that SPRLT, LKSSAN, and SSTN have better results compared to other comparison algorithms in the IP dataset. By contrast, OA and Kappa of ESNet have better and more robust performance. In the PU and UH datasets, ESNet presents a remarkable advantage, with the

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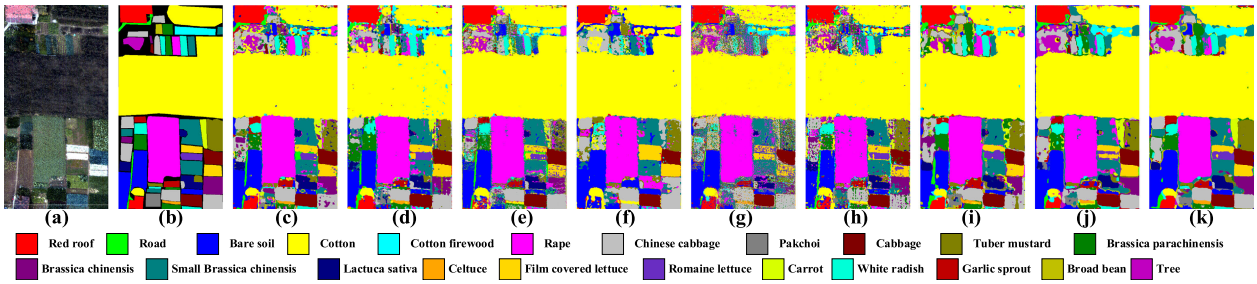


Fig. 19. Visualization of the classification maps for the WH dataset. (a) True-color image of WH. (b) Ground truth of WH. (c) SSTN. (d) SSFTT. (e) HiT. (f) SPRLT. (g) ASSMN. (h) A2S2K. (i) FDGC. (j) LKSSAN. (k) ESNet.

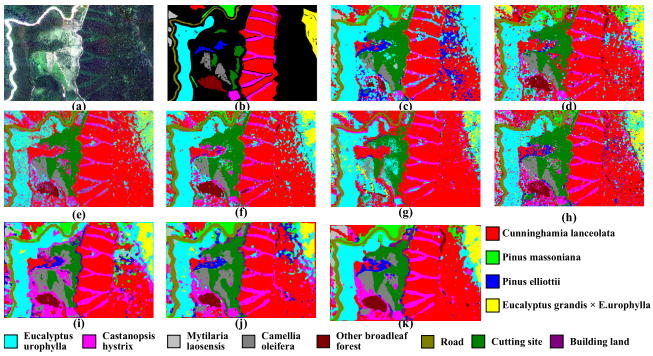


Fig. 20. Visualization of the classification maps for the TREE dataset. (a) True-color image of TREE. (b) Ground truth of TREE. (c) SSTN. (d) SSFTT. (e) HiT. (f) SPRLT. (g) ASSMN. (h) A2S2K. (i) FDGC. (j) LKSSAN. (k) ESNet.

mean values of its metrics outperforming those of the other compared algorithms, while its standard deviation achieves optimal versus suboptimal results several times. Although the AA of ESNet in the IP dataset is lower than that of SPRLT, the overall performance of ESNet demonstrates its superior performance in the spatially disjoint dataset.

Tables IV–VI show the PA, OA, AA, and Kappa of each model in few-shot datasets. As shown in Tables IV–VI, we can find that ESNet obtains optimal OA, AA, kappa, and the most optimal PA. In the UH2018 dataset, the number of training samples for classes 3, 7, and 17 is 7, 3, and 2, respectively, but ESNet still has a better performance in the recognition of these three categories. In the WH dataset, the GCN-based FDGC and CNN-based LKSSAN have the better OA, AA, and kappa among the compared algorithms, but are still lower than the corresponding values of ESNet. In the TREE dataset, the training samples for categories 7, 8, and 12 are insufficient, which can effectively estimate the ESNet with few-shot samples. As shown in Table VI, although the recognition accuracy of ESNet for class 12 is poor, it is higher in classes 7 and 8. These three datasets have different spatial resolutions and unbalanced and insufficient samples, which pose a serious challenge for existing SOTA algorithms. In contrast, ESNet obtains excellent and robust results in these datasets, which amply demonstrates the advantages of the anti-hierarchical encoder–decoder network proposed for HSI classification.

The results of McNemar's test between ESNet and other SOTA methods are shown in Table VII. Since the results of HiT and ASSMN are uncompetitive, we did not analyze these results further in this part. We can see that in six datasets, besides the results of SPRLT in the IP dataset, the improvement of performance produced by ESNet is statistically significant as compared with other methods. Moreover, we also can deduce that ESNet outperforms SPRLT in the IP dataset based on the formulation of McNemar's test.

2) Qualitative Evaluation: Figs. 15–20 show the classification results of the models in the six datasets. We can see that the classification maps of SSTN, SSFTT, SPRLT, FDGC, and ESNet have good performance on the IP dataset. There is a cloud-shadowing region in the UH2013 dataset. As shown in Fig. 17, all algorithms are affected by shadows, but SSTN and SSFTT have better spatial structures in this region. In the PU dataset, the ESNet correctly labels easily confused classes, such as Meadows and Bare soil, which is marked with black and white rectangles. ESNet can obtain competitive results in the spatial disjoint datasets and even has more excellent performance in some easily confused classes, which demonstrate the reliable capability and prospect of the anti-hierarchical encoder–decoder framework.

As shown in Fig. 18, the classification maps of SSTN, HiT, SPRLT, and A2S2K contain salt and pepper noise because these models fail to extract large-scale spatial context. Although FDGC and LKSSAN can obtain smooth images, the edge-keeping effect of these methods is poor. In contrast, ESNet has a good result in the recognition of fine classes and large-scale buildings, which indirectly reflect that the differentiated treatment of the center and surrounding features can effectively improve the feature recognition capability of the model.

According to Fig. 19(a) and (b), we can see that the distribution area and intraclass variation of Cotton are huge in the WH dataset. Therefore, A2S2K, SSFTT, and ASSMN have more serious misclassifications in the distribution area of Cotton. In addition, there are many kinds of features above and below in the WH dataset, and their distribution is fragmented, which is employed to evaluate the ability to detail reconstruction. ESNet has an excellent performance in recognizing the features in these two areas. The experimental results reflect that ESNet can effectively exploit large- and small-scale feature details to improve classification accuracy, which demonstrates the importance of fine-grained extraction of spatial features for the reconstruction of multiscale spatial structures.

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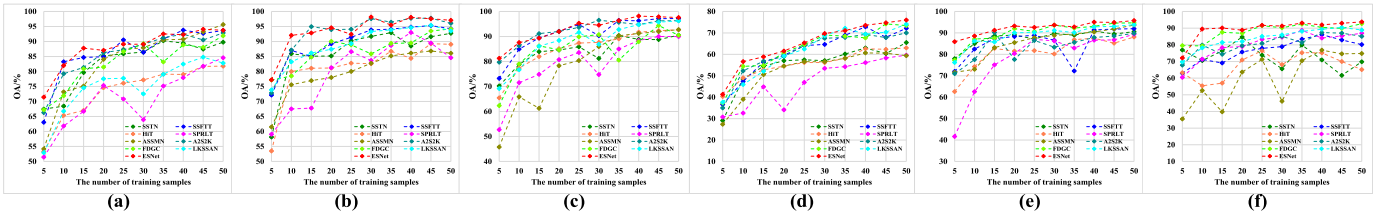


Fig. 21. Effect of the numbers of training samples for SSTN, SSFTT, HiT, SPRLT, ASSMN, A2S2K, FDGC, LKSSAN, and ESNet on (a) IP, (b) PU, (c) UH2018, (d) UH2018, (e) WH, and (f) TREE datasets.

TABLE III

OAs, AAs (%), and Kappa Coefficients Achieved by Different Methods on the IP, PU, and UH2013 Dataset. The Best Record in Each Row Is Bolded and the Suboptimal Results Are Underlined

Data sets	Metrics	Transformer-based				RNN, GCN and CNN-based				ESNet
		SSTN	SSFTT	HiT	SPRLT	ASSMN	A2S2K	FDGC	LKSSAN	
IP	OA (%)	86.95±1.01	81.51±2.25	80.67±2.50	87.90±0.58	81.92±2.34	86.46±1.56	77.25±1.94	87.55±0.51	**88.95±0.29**
	AA (%)	75.45±1.20	72.97±2.87	76.47±2.88	**83.15±0.72**	72.65±1.98	74.90±1.27	64.86±3.58	80.41±0.84	81.10±1.32
	Kappa*100	85.19±1.1	79.01±2.6	77.98±2.9	86.24±0.7	79.20±2.7	84.63±1.8	74.12±2.2	85.82±0.6	**87.48±0.3**
PU	OA (%)	86.63±9.68	83.37±1.67	88.54±1.15	94.35±**0.86**	86.21±1.66	85.62±2.14	77.14±3.71	91.77±0.99	**94.73**±1.22
	AA (%)	89.21±5.03	87.51±**0.37**	86.38±1.46	92.89±0.64	84.69±1.73	89.71±0.79	80.44±2.65	92.22±0.83	**93.49**±0.47
	Kappa*100	82.38±12	78.60±2.2	84.68±1.5	92.44±**1.1**	81.29±2.2	81.52±2.6	71.20±4.1	88.72±1.4	**92.80**±1.7
UH2013	OA (%)	87.33±1.52	87.94±0.72	79.86±0.72	84.72±1.36	67.04±1.97	84.26±1.06	77.18±4.44	83.44±1.79	**88.71±0.44**
	AA (%)	88.63±2.07	91.08±**0.80**	81.83±1.13	88.32±1.43	66.44±2.27	87.12±1.16	80.93±3.69	86.86±1.92	**91.76**±1.93
	Kappa*100	86.25±1.6	86.90±0.7	78.25±0.8	83.49±1.5	64.29±2.1	83.02±1.1	75.23±4.8	82.02±1.9	**87.75±0.5**

TABLE IV

PAs, OAs, AAs (%), and Kappa Coefficients Achieved by Different Methods on the UH2018 Dataset. The Best Record in Each Row Is Bolded and the Suboptimal Results Are Underlined

Class No.	Transformer-based				RNN, GCN and CNN-based				ESNet
	SSTN	SSFTT	HiT	SPRLT	ASSMN	A2S2K	FDGC	LKSSAN	
1	**87.41**±4.2	78.09±**0.9**	**86.2**±3.48	83.6±10.79	83.86±6.04	84.83±1.09	73.19±2.81	81.36±2.26	79.74±2.21
2	87.46±1.7	90.66±0.76	**94.05**±1.03	93.36±5.21	**95.55**±2.21	93.17±**0.44**	91.87±1.15	90.01±0.66	91.71±0.72
3	98.1±1.4	97.13±1.65	96.71±2.05	94.42±8.18	87±14.36	99.3±0.29	96.83±1.57	98.76±2.25	**100±0**
4	84.63±5.54	92.3±1.73	94.56±1.06	91.35±4.85	**96.33**±0.96	93.64±**0.54**	92.11±1.8	94.86±0.56	95.72±0.75
5	76.61±7.3	81.89±5.01	80.02±4.54	**91.77**±2.45	76.02±4.19	87.07±**1.23**	78.47±4.77	87.96±1.89	89.63±2.54
6	91.35±5.28	98.24±0.73	87.83±3.1	92.97±6.42	99.05±0.67	92.52±1.93	98.83±1.88	**99.41**±0.47	99.35±**0.31**
7	81.15±5.09	95.92±3.59	**100±0**	0±0	48.85±3.98	**100±0**	50.38±8.15	**100±0**	98.84±1.22
8	82.27±4.42	94.5±1.38	87.65±1.76	86.15±5.67	94.94±1.39	93.79±0.97	95.39±1.31	**95.73**±0.63	**97.78±0.32**
9	95.76±0.19	97.49±0.44	95.07±0.27	95.04±2.36	98.41±0.24	97.36±0.2	**98.46**±0.19	97.85±0.19	98.05±**0.15**
10	70.3±**0.97**	82.92±1.14	74.69±1.43	71.52±5.4	80.21±1.47	79.12±1.65	78.25±2.27	85.5±1.22	**87.02±1.06**
11	69.15±3.23	75.41±2.84	71.86±1.16	70.62±5.33	67.84±1.88	**81.06±0.66**	61.95±2.37	78.52±1.08	79.88±1.27
12	0±0	35.15±4.61	37.72±3.1	43±38.68	9.71±1.54	**47.31±0.8**	0.48±0.96	36.41±1.42	32.16±4.67
13	66.14±4.99	86.98±0.8	78.84±2.26	81.62±3.26	90.43±5.77	87.04±**0.49**	**92.01**±1.31	88.48±1.68	90.92±1.17
14	87.25±2.66	87.82±2.31	89.74±1.29	90.78±1.98	**98.26±0.58**	90.51±1.21	92.19±2.89	93.28±1.26	97.39±0.97
15	82.72±7.68	98.39±0.42	93.7±1.61	97.14±1.22	97.28±1.06	97.51±**0.34**	90.32±1.54	95.39±0.98	**99.29±0.51**
16	77.28±3.68	92.76±**0.55**	88.04±1	89.55±2.46	91.56±1.96	92.99±1.28	86.09±4.1	95.57±0.68	**97.09±1.22**
17	85.19±3.62	86.99±2.45	91.07±2.92	0±0	37.08±8.53	95.2±3.51	0±0	93.64±8.05	**98.69±0.95**
18	91.53±2.36	91.17±2.92	85.15±2.01	87.64±6.17	93.73±2.2	92.35±1.98	74.95±6.63	97.22±**0.83**	**97.3±0.83**
19	94.99±2.54	92.82±1.47	95.07±1.48	90.23±5.21	97.27±2.67	98.3±0.45	91.81±1.8	98.89±0.47	**99.69±0.42**
20	94.88±3.93	98.77±0.63	98.08±0.95	94.5±3.4	97.75±1.34	98.42±0.9	99.22±0.47	99.39±0.4	**99.86±0.03**
OA (%)	85.69±0.81	92.02±0.15	88.88±0.23	88.18±0.85	92.49±0.70	92.35±0.21	90.96±0.97	**93.36±0.14**	**94.30±** 0.17
AA (%)	80.21±2.46	87.77±0.61	86.30±0.37	77.26±1.41	82.06±1.94	90.07±**0.14**	77.14±2.49	90.41±0.55	**91.51±** 0.34
Kappa*100	81.28±1.0	89.56±0.2	85.42±0.3	84.50±1.2	90.14±0.9	90.0±0.3	88.16±1.3	91.32±0.2	**92.55±0.2**

According to Fig. 20, we can see that most of the comparison algorithms have severe speckle phenomena, but ASSMN, FDGC, LKSSAN, and ESNet effectively alleviate this problem. Moreover, the classification map of ESNet not only can reflect the spatial distribution in the center part of the TREE dataset but also has accurate edges. In addition, we can find that *Eucalyptus grandis* × *Eucalyptus urophylla* is more difficult to classify, and all the comparative algorithms have serious misclassifications in this category, but ESNet can recognize category four with high accuracy. Moreover, ESNet can effectively alleviate the central empty hole in the recognition of parcels with different sizes, which indirectly

TABLE V

PAs, OAs, AAs (%), and Kappa Coefficients Achieved by Different Methods on the WH Dataset. The Best Record in Each Row Is Bolded and the Suboptimal Results Are Underlined

Class No.	Transformer-based				RNN, GCN and CNN-based				ESNet
	SSTN	SSFTT	HiT	SPRLT	ASSMN	A2S2K	FDGC	LKSSAN	
1	90.57±5.33	94.78±1.42	93.14+2.7	87.19+5.39	87.15±4.95	95.73±2.43	94.67+1.28	97.37+1.01	**98.11+0.47**
2	67.33+**3.88**	63.86±15.78	60.41±8.75	38.82±33.04	64.29+7.63	66.49±6.41	66.77±4.37	**72.2**±10.54	66.65±4.80
3	84.71±4.01	81.15±6.11	74.78±8.02	70.04±8.43	**90.5**±2.08	82.21±2.41	87.13±**0.44**	81.42±1.18	86.00±2.78
4	97.54±0.27	97.05±1.66	94.76±1.19	92.35±1.61	97.5±2.40	95.9±0.86	**99.53**±0.26	98.92±0.27	99.19±**0.16**
5	87.48±6.98	74.15±9.82	54.99±7.54	21.45±28.28	67.13±12.72	59.17±10.15	65.8±9.44	87.06±5.54	**94.80±1.44**
6	93.31±2.53	85.58±6.26	84.08±4.97	77.41±7.4	90±3.50	91.33±1.39	98.4±**0.33**	93.55±3.28	**98.87±**0.77
7	77.41±2.36	86.16±8.21	72.63±11.24	57.61±3.19	73.29±3.86	74.74±5.75	**87.69**±4.22	84.47±1.64	84.01±1.56
8	50.52±**1.68**	66.57±8.799	20.2±8.43	15.85±21.78	5.91±1.84	29.76±7.36	63.23±3.9	67.53±10.18	**73.15±**6.10
9	95.36±5.03	95.69±0.957	89.47±8.28	71.1±12.78	78.4±7.92	**96.36**±1.51	94.35±1.28	93.67±4.31	96.29±1.12
10	79.8±8.98	83.42±8.63	55.26±4.13	70.32±10.99	49.8±7.75	79.57±2.29	75.02±**0.7**	81.49±8.5	**89.54±**3.26
11	75.65±7.03	63.12±11.36	54.35±7.31	60.36±11.07	19.7±7.58	67.15±7.06	**84.48±2.16**	77.21±4.47	83.09±3.57
12	63.78±5.32	**67.42**±18.05	44.01±28.37	54.59±26.16	37.43±12.86	55.74±5.08	60.19±**1.73**	63.53±5.74	**87.78±**4.44
13	80.1±4.65	68.55±4.34	66.11±1.44	58.84±**0.93**	69.22±3.46	70.14±3.14	74.17±3.89	74.36±3.02	**81.24±**2.80
14	84.17±**3.08**	61.87±16.06	53.15±20.47	27.85±37.44	52.16±8.37	90.3±3.14	80.81±5.51	86.18±4.2	**94.28±**4.96
15	**97.81±3.39**	79.01±10.65	58.43±23.09	0±0	14.96±15.01	70.08±23.26	71.37±7.98	75.56±9.92	83.30±3.64
16	**96.95±2.23**	87.98±4.93	73.13±5.94	61.08±3.16	74.22±9.05	82.56±3.11	87.27±2.43	88.46±6.19	95.64±3.14
17	69.32±9.26	**86.67**±4.35	35.57±9.8	8.39±16.77	37.29±18.41	66.81±11.97	79.84±**0.39**	70.56±8.47	67.61±3.13
18	69.78±8.04	76.54±21.81	54.52±5.98	0±0	15.91±10.59	61.14±10.43	65.86±17.34	71.62±15.01	**89.43±2.73**
19	**94.81±1.89**	82.54±7.58	76.01±9.95	68.11±13.79	40.1±6.391	91.34±3.66	89.34±1.21	85.1±5.76	90.68±5.29
20	76.38±9.81	65.57±13.19	71.78±19.32	**89.27**±13.86	37.51±21.19	77.67±12.66	67.15±11.79	61.16±**4.11**	73.25±5.00
21	11.14±6.76	64.65±19.74	2.57±**1.63**	0±0	6.58±6.593	23.6±7.06	**75.19**±5.92	26.21±8.39	17.41±2.53
22	74.07±6.54	70.14±14.34	63.64±8.11	20±40	21.48±7.01	58.63±7.93	54.54±10.02	80.12±10.28	**86.37±3.95**
OA (%)	89.36±0.34	86.32±1.610	81.64±2.78	78.89±1.08	79.69±2.13	86.30±0.41	90.65±0.63	89.82±0.48	**93.06±0.52**
AA (%)	78.09±0.62	77.38±1.687	61.50±6.10	61.50±2.58	51.39±3.45	72.11±0.2.00	78.31±1.35	78.08±1.29	**83.49±1.39**
Kappa*100	86.49±0.4	82.60±2.1	76.47±3.6	72.67±1.3	74.20±2.75	82.54±0.6	88.14±0.8	87.13±0.6	**91.22±0.7**

TABLE VI

PAs, OAs, AAs (%), and Kappa Coefficients Achieved by Different Methods on the TREE Dataset. The Best Record in Each Row Is Bolded and the Suboptimal Results Are Underlined

Class No.	Transformer-based				RNN, GCN and CNN-based				ESNet
	SSTN	SSFTT	HiT	SPRLT	ASSMN	A2S2K	FDGC	LKSSAN	
1	86.77±3.53	88.66±1.81	81.86±1.82	85±8.11	**96.56**±1.67	90.95±0.88	95.31±1.03	94.14±0.96	95.05±0.69
2	83.02±1.15	81.79±4.46	74.23±12.93	86.74±7.08	5.24±**2.48**	79.2±5.74	84.55±5.55	84.29±9.72	**90.79±**3.61
3	**79.59±**1.09	85.57±4.61	38.38±7.29	87.08±**0.39**	7.41±2.19	73.1±8.48	75.59±3.76	66.46±9.89	75.86±6.71
4	20±40	81.72±2.25	57.83±9.65	93±6.58	36.66±5.25	93.29±4.78	83.78±5.22	83.26±4.34	**99.59±0.47**
5	57.91±3.22	79.8±2.41	65.55±7.12	81.72±5.9	88.16±3.25	84.18±2.1	**96.59±0.86**	85.56±1.47	94.42±2.47
6	**88.06±**6.95	74.26±2.34	59.53±9.13	75.64±2.33	66.76±4.47	75.92±2.54	80.39±**1.67**	80.84±2.62	82.39±2.24
7	57.15±22.5	**94±2.11**	83.53±7.44	82.2±26.33	18.53±9.1	76.15±9.77	71.82±14.56	75.42±2.87	85.50±4.30
8	93.68±7.78	94.09±1.2	93.29±**0.88**	88.25±6.07	39.17±6.87	94.3±2.9	**94.39**±2.83	93.96±2.36	**99.13±**1.06
9	**99.31±**1.24	78±2.95	75.7±6.11	69.56±7.34	22.78±2.77	81.18±7.5	75.74±4.38	93.58±2.41	96.34±**0.80**
10	93±5.56	96.66±1.3	98.97±**0.13**	93.74±3.36	80.15±8.44	98.48±0.69	88.9±1.01	99.04±0.9	**99.78±**0.18
11	88.86±3.89	90.15±2.89	88.53±2.48	82.53±7.57	83±5.22	94.62±1.41	**99.61±0.27**	88.6±3.76	96.08±2.05
12	49.05±3.95	**54.19**±22.93	22.78±8.15	0±0	50.45±6.15	16.75±5.9	31.64±2.39	8.57±**2.04**	10.8±4.01
OA (%)	77.30±2.87	84.62±0.63	75.92±4.11	82.04±5.03	74.71±1.33	87.19±0.82	90.84±1.06	88.49±0.52	**93.09±0.07**
AA (%)	74.70±5.82	**83.24**±1.85	70.01±5.16	77.12±3.66	49.57±1.21	79.84±2.56	81.53±1.74	79.48±0.88	**85.48±1.05**
Kappa*100	70.39±3.8	80.36±0.9	68.52±5.6	76.73±6.9	66.85±1.6	83.68±1.0	**88.38**±1.3	85.45±**0.7**	**91.29±0.7**

prove that the adequate expression of the center features can better exploit the intrinsic representation of the center pixel of the patch.

3) Number of Samples Sensitivity Analysis: As displayed in Fig. 21, the accuracy of ESNet increases smoothly with the increase of the number of samples, the OA of ESNet is generally superior to the other comparison algorithms in the six datasets, and the accuracy of ESNet is only slightly

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TABLE VII

RESULTS OF MCNEMAR'S TEST BETWEEN ESNET AND OTHER SOTA METHODS. THE BEST RECORD IN EACH ROW IS BOLDED AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Dataset	SSTN	SSFTT	SPRLT	A2S2K	FDGC	LKSSAN
IP	5.53	13.73	**1.43**	4.41	17.07	3.94
PU	4.74	57.95	64.23	22.94	61.86	**3.84**
UH2013	14.79	**4.65**	18.09	22.28	31.49	21.64
UH2018	177.34	62.50	125.08	47.36	131.86	**33.87**
WH	42.08	65.33	188.97	99.46	69.62	**41.90**
TREE	152.65	108.43	103.12	82.55	**50.39**	82.11

TABLE VIII

TRAINING TIME, TEST TIME, AND PARAMETER SIZE ACHIEVED BY DIFFERENT METHODS ON THE UH2018, WH, AND TREE DATASETS. THE BEST RECORD IN EACH ROW IS BOLDED AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Data sets	Metrics	Transformer-based				RNN, GCN and CNN-based				ESNet
		SSTN	SSFTT	HiT	SPRLT	ASSMN	A2S2K	FDGC	LKSSAN	
UH2018	Train (s)	44.13	5.65	102.64	35.10	**4.16**	54.51	9.88	53.58	61.24
	Test (s)	1680.29	504.34	3717.59	2462.88	**454.27**	2548.29	639.52	3653.81	1633.31
	Parameter	**0.014M**	0.15M	53.16M	0.83M	2.62M	0.14M	2.71M	0.75M	1.28M
WH	Train (s)	3.41	**0.44**	5.70	3.01	3.06	27.97	0.62	43.45	2.65
	Test (s)	1354.51	332.68	2745.52	2036.39	388.41	2517.20	**228.73**	2719.08	1254.05
	Parameter	**0.025M**	0.15M	68.81M	0.86M	2.99M	0.48M	2.85M	0.75M	1.28M
TREE	Train (s)	1.99	**0.43**	3.23	1.59	1.93	4.08	0.78	2.42	1.75
	Test (s)	743.45	335.91	1700.67	1043.85	253.30	1151.48	**202.11**	1503.06	581.74
	Parameter	**0.017M**	0.15M	55.42M	0.84M	2.74M	0.26M	2.19M	0.75M	1.29M

lower than the other algorithms at a certain number of training samples. Although the OAs of SSFTT and SSTN are better than the OA of ESNet in some instances, the SSFTT and SSTN cannot get imposing results in spatial disjoint datasets. In addition, the performance of ESNet in the six datasets is always less sensitive to sample numbers. In summary, ESNet achieves optimal results compared to the SOTA algorithms in different scenarios, scales, tasks, and training samples.

4) Complexity Analysis: Table VIII shows the training time for training one epoch, the total testing time, and the number of parameters of the model in UH2018, WH, and TREE datasets with larger volumes of data. As we can see, SSTN has the smallest parameter size, and SSFTT spends the least time on training and testing. Besides, A2S2K has only a 0.14M parameter size but spends 2548.29 s to test the model in UH2018, which reflects that a lighter model does not mean less time spent. According to the results shown in Tables IV–VII, we can speculate that among the comparison algorithms, LKSSAN has the best performance, but ESNet spends less time on prediction. The training and testing time of ESNet is more expensive, which may be because there are many self-operations based on the feature map when calculating the long- and short-range 3-D weights. Therefore, in the future, we can further simplify the internal operations of the ESNet to push the model further on speed.

V. DISCUSSION

In this section, we discuss the effectiveness of the ESNet core modules, including SRM, SSAM, and DSFM. In addition, the effect of Gaussian noise on the model performance.

TABLE IX

OAS (%) ACHIEVED BY DIFFERENT STRUCTURES IN THE UH2018, WH, TREE, IP, PU, AND UH2013 DATASETS. THE BEST RECORD IN EACH ROW IS BOLDED

SRM	SSAM	DSFM	Overall Accuracy (%)					
			UH2018	WH	TREE	IP	PU	UH2013
-	-	-	91.50±0.11	91.35±**0.12**	90.53±0.44	81.08±**0.03**	86.48±**0.79**	85.50±0.97
✓	-	-	93.62±**0.06**	92.03±0.21	92.66±1.11	86.55±0.43	91.26±1.71	86.54±**0.43**
✓	✓	-	94.10±0.09	92.99±**0.12**	92.74±0.47	86.72±1.23	92.20±1.70	86.72±1.23
✓	-	✓	93.86±0.11	92.21±0.19	93.07±0.44	87.81±1.38	92.34±1.68	87.80±1.38
✓	✓	✓	**94.30**±0.17	**93.06**±0.52	**93.09**±**0.07**	**88.45**±1.86	**94.73**±1.22	**88.71**±0.44

A. Ablation Analysis of ESNet

In this section, we investigate the performance gain of the proposed ESNet by adding (in the stepwise fashion) different modules (i.e., SRM, SSAM, and DSFM) in networks. We conducted extensive ablation experiments on six datasets, and the results of each experiment are shown in Table IX.

Unlike the traditional scale change module, SRM achieves scale change by controlling edge pixel expansion and stripping, which enables ESNet to exploit spatial information with different geospatial scales at different stages of the encoder and decoder. As shown in Table IX, SRM can significantly improve the model accuracy from 0.68% to 5.47%, which demonstrates the significance of exploiting refined spatial information in patch-based HSI classification methods. Combining this experimental result with the results of the hyperparameter experimental about scale factor, we can confirm the reasonableness of the research motivation, that is, it is crucial that the refined extraction of central and surrounding features, and the central features play a more decisive role than the surrounding features in the classification task.

The SSAM can adaptively extract the spectral and spatial features by a novel dual-branch attention. Hence, we can see that SRM-based and the SRM + DSFM-based models can obtain excellent results after adding the SSAM whether in scenes with high spatial heterogeneity or spectral homogeneity. Although SSAM has less improvement in model performance compared to SRM, it still demonstrates the effectiveness of SSAM in assisting ESNet in extracting features.

To fuse deep and shallow features in a patch-based encoder–decoder framework, we propose the DSFM that can fuse deep and shallow features without destroying the spectral sequence of the 3-D patch. As a result, we can see from Table IX that the DSFM can improve the performance of SRM-based and SRM + SSAM-based models. Note that when DSFM is not used, we adopt the traditional skip connection for deep and shallow feature fusion. Hence, we can induce that preserving spectral sequences in HSI datasets can improve model performance. Interestingly, DSFM can achieve superior results in urban scenes compared to vegetated scenes. This may indicate that spectral information is more necessary for scenes with complex spatial structures. In other words, preserving spectral sequence information is more helpful for mining complex spatial information, which complies with the

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TABLE X
OAS, AA (%), AND KAPPA ACHIEVED BY DIFFERENT STRUCTURES IN THE WH, TREE DATASETS UNDER GAUSSIAN NOISE DEGRADATION. THE BEST RECORD IN EACH ROW IS BOLDED

Noisy		0	0.005	0.01	0.02	0.03	0.04
WH dataset	OA/%	**93.23**	90.62	90.18	92.13	86.68	87.26
	AA/%	**83.44**	79.87	78.59	82.40	72.45	72.19
	Kappa	**0.9144**	0.8813	0.8756	0.9003	0.8312	0.8386
TREE dataset	OA/%	**92.74**	88.19	83.96	76.99	76.7	70.97
	AA/%	**87.01**	71.42	68.95	64.76	60.71	55.78
	Kappa	**0.9081**	0.8313	0.7743	0.6918	0.6771	0.6229

hypothesis that spectral and spatial features are complementary. WH and TREE datasets have similar spatial resolution and scenes, but DSFM can obtain better results in the TREE dataset with a more complex spatial structure, which further supports the above hypothesis.

B. Noise Analysis

In practical application, HSI is generally affected by noise. Based on the qualitative analysis of the ESNet, we can see that the algorithm is more seriously affected by noise in the WH and TREE datasets. Therefore, in this section, we analyze the Gaussian noise immunity of ESNet by conducting a series of experiments in the WH and TREE datasets. The mean value is fixed as 0, and the variance of Gaussian noise is adjusted to 0.001, 0.005, 0.01, 0.02, 0.03, and 0.04.

As shown in Table X, the model accuracy shows a decreasing trend with increasing noise. In the dataset TREE, the effect of noise on ESNet is the most serious. When the variance of Gaussian noise is 0.04, it causes the OA to decrease by 5.97% and 21.77% in the WH and TREE datasets, respectively. Moreover, the WH dataset represents data from farmland scenes and the TREE dataset reflects the forest scene, the spatial structure of the WH dataset is more regular than that of the TREE dataset. Hence, we can guess that the spatial information may have a role in improving the noise immunity of the model.

Noise, as a key environmental factor, is shown in this experiment to have a serious impact on model accuracy. Therefore, we will research how to improve the ability of ESNet to resist noise in the future for high-precision HSI classification.

VI. CONCLUSION

In HSI classification, both central and surrounding features are crucial, and the central features play a more decisive role than the surrounding features. Although many existing frameworks have been used to extract spatial information, there is still no framework dedicated to exploiting the center and surrounding information. To resolve the problems, we propose an anti-hierarchical encoder–decoder framework (ESNet), which utilizes SRM, SSAM, and DSFM to achieve high-precision classification tasks in six datasets containing city, agriculture, and forest scenarios.

SRM, by combining with the encoder–decoder structures, purposefully exploits surrounding and center information, allowing both information to play to their respective strengths while taking full advantage of their intrinsic associations. SSAM can explore, enhance, and integrate spectral, spatial, and spectral–spatial information by utilizing long- and short-range spectral–spatial attention mechanisms. DSFM employs a lightweight structure based on the deep and shallow characteristics of HSI to achieve a scientific fusion of abstract and detailed information without destroying the spectral sequence. Comprehensive experiments and ablation studies in six datasets have demonstrated the effectiveness and superiority of ESNet. However, both the weak noise immunity and slow prediction speed of ESNet limit its application in the real world. In the future, we will take the intrinsic physical properties of HSI as a starting point to research the model that is more resistant to noise through lightweight construction.

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Zhaojie Pan received the B.Sc. and M.Sc. degrees from China University of Petroleum (East China), Qingdao, China, in 2021 and 2024, respectively. He is currently pursuing the Ph.D. degree with the School of Mathematics, Southeast University, Nanjing, China. He is also a joint Ph.D. student with the School of Information and Control Engineering, Qingdao University of Technology, Qingdao. His research interests include hyperspectral image classification and deep learning.

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Genyun Sun (Senior Member, IEEE) received the B.Sc. degree from Wuhan University, Wuhan, China, in 2003, and the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2008.

He is currently a Professor with China University of Petroleum (East China), Qingdao, China. He is also with the Laboratory for Marine Mineral Resources, Qingdao National Laboratory for Marine Science and Technology, Qingdao. His research interests cover remote sensing image processing, including hyperspectral and high-resolution remote sensing, and intelligent optimization.



Ziyan Ling received the B.Sc. degree from Nanjing Forestry University, Nanjing, China, in 2008, and the M.Sc. and Ph.D. degrees from Beijing Normal University, Beijing, China, in 2011 and 2024, respectively.

She is currently an Associate Professor with the School of Information and Control Engineering, Qingdao University of Technology, Qingdao, China. Her research interests include artificial intelligence, urban remote sensing, and wetland remote sensing.



Aizhu Zhang (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from China University of Petroleum (East China), Qingdao, China, in 2011, 2014, and 2017, respectively.

She is currently an Associate Professor with China University of Petroleum (East China). She is also with the Key Laboratory of Poyang Lake Wetland and Watershed Research, Ministry of Education, Jiangxi Normal University, Nanchang, China. Her research interests include multisource remote sensing processing, wetland remote sensing, and city environment remote sensing.



Hang Fu received the B.Sc. degree in geomatics engineering from China University of Petroleum (East China), Qingdao, China, in 2019, where he is currently pursuing the Ph.D. degree in computer technology and resource information engineering.

His research interests are related to hyperspectral image (HSI) feature extraction and classification.

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